

PROGRAMME: BCA (Bachelor of Computer Applications)- CTIS SEMESTER – VI

Teaching Learning and Evaluation Plan

Course Information:

Course Code: **23BCAC6C02**

Course Title: **WINDOWS AZURE WEB SERVICES**

Credits Units: **03**

Contact Hours: **45**

L-T-P-E: **2-0-0-3**

IA: ESE Weightage - **50: 50** Pass Marks (IA & ESE) **0:40(MIN ESE 18)** Aggregate Pass Marks: **40**

ESE Question Paper Marks: **50**

Special Examination Fees: **NA**

Pre-requisite (if any): **NA**

Course Facilitator (s): **Neha Singh, Assistant Professor, School of CS and IT**

Programme Outcomes (POs)

At the end of the programme, students will be able to

PO 1	Engineering Knowledge - Apply knowledge of computing, mathematics, science, and engineering fundamentals to cloud computing solutions.
PO 2	Problem Analysis - Identify, formulate, and analyze complex computing problems related to cloud services and Azure infrastructure.
PO 3	Design/Development of Solutions - Design and develop cloud-based solutions using Azure services that meet specified needs.
PO 4	Conduct Investigations - Use research-based knowledge and methods to investigate complex Azure cloud computing problems.
PO 5	Modern Tool Usage - Create, select, and apply appropriate techniques and modern tools including Azure Portal, Visual Studio for cloud application development.
PO 6	Professional Ethics - Apply ethical principles and commit to professional ethics in Azure cloud service deployment and management.
PO 7	Life-long Learning - Recognize the need for and have the ability to engage in independent and life-long learning in cloud technologies.
PO 8	Individual and Team Work - Function effectively as an individual and as a member in diverse teams in cloud computing projects.
PO 9	Communication - Communicate effectively on complex cloud computing activities through reports, presentations, and documentation.
PO 10	Project Management - Demonstrate knowledge and understanding of project management principles in Azure deployments.
PO11	Ethics and Social Responsibility - Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal issues in cloud computing.
PO12	Environment and Sustainability - Understand the impact of cloud computing solutions in societal and environmental contexts, and demonstrate knowledge of sustainable cloud practices including resource optimization, energy-efficient computing, and responsible use of cloud resources.

Program Specific Outcomes (PSO's)

PSO 1	Cloud Computing Proficiency Demonstrate proficiency in designing, implementing, and managing cloud-based solutions using Microsoft Azure platform, including compute, storage, networking, and database services to solve real-world business problems.
PSO 2	Cloud Infrastructure Management Apply knowledge of cloud infrastructure components to architect, deploy, and manage scalable, secure, and cost-effective Azure solutions including virtual machines, storage accounts, virtual networks, and load balancers for enterprise applications.

PSO 3	Cloud Security and Identity Management Implement security best practices and identity management solutions using Azure Active Directory, Network Security Groups, and multi-factor authentication to ensure secure access and data protection in cloud environments.
PSO 4	Cloud Application Development and Deployment Design, develop, and deploy web applications and databases on Azure platform using modern development tools and DevOps practices, with emphasis on automation, monitoring, optimization, and continuous improvement of cloud services.

Course Objectives (COB's)

COB1	To understand the fundamentals of Cloud Computing and basics of Windows Azure
COB2	To apply the skills in creating Windows Azure Storage
COB3	To apply the knowledge in developing Windows Azure Networking
COB4	To analyse the Azure Active Directory services with implementation
COB5	To create Azure website for deploying credentials

Course Outcomes (CO's)

Sl No	Course Outcome	Description	Bloom's Taxonomy Level
1	CO 1	Describe the basics of cloud services in Windows Azure like computer service, network service, data service and App service.	L2
2	CO 2	Explain the subscription and pricing of Azure services	L4
3	CO 3	Use Azure Storage services	L3
4	CO 4	Create Network infrastructure using Azure services	L2
5	CO 5	Use Azure Active Directory and Database services.	L5

CO-PO/PSO Mapping:

Cours e Outco me	Blooms Taxono my Levels	Program Outcomes (PO's) Mapped												Program Specific Outcomes (PSO's)			
		P O1	P O2	PO 3	P O 4	P O 5	P O 6	P O7	P O 8	P O 9	PO 10	PO 11	P O1 2	PS O1	PSO 2	PS O3	PS O4
														L3	L4	L5	L6
C01	L2	3	2	-		-	-	2	-	2	-	-	-	3	1	1	1
C02	L3	3	2	2	3	3	-	3	-	2	-	-	-	3	2	2	2
C03	L4	3	3	3	3	3	-	3	-	2	2	-	-	3	3	2	2
C04	L3,L4	3	3	2	3	3	3	3	2	2	3	2	2	3	3	3	3
C05	L5	3	3	3	3	3	3	3	2	2	2	3	2	3	3	3	3
CO Avg.		3	2.6	2.8	3	3	3	3	2	2	2.3	2	2	3	2.4	2.2	2.2

Course Contents:

Module	Details	Contact Hours
I	Getting Started with Azure Overview of Cloud Computing – Various Cloud Offerings – Azure Basics – Azure Services – Azure Portals – Preview Portal, Management Portal, Subscription Management – Billing – Pricing Calculator - Azure Virtual Machines: Virtual Machine (VM) Basics – Status, IP Address, Creating and Configuring Virtual Machines – Configuring VM disks – Virtual Machine Management.	9
II	Azure Storage Storage Basics – Storage Types – Azure Storage Offerings – Understanding Azure Regions – Using Storage Accounts – Enabling Larger and Faster Storage – Resizing Azure Disks – Using Premium Storage – Monitoring Azure Storage Accounts – Best Practices for Azure Storage – Azure VM Storage Types – Azure Files – Managing Azure Storage.	9
III	Azure Networking Basics of Virtual Networks –Address Spaces, Subnets, DNS Servers – Creating and Using Virtual Networks – Network Security Groups – Virtual Appliances – Load Balancer basics – Configuring Load Balancers – Creating and Using Load balancers – Azure VPN - Windows Azure Network Services - Virtual Network- Traffic Manager - Using Traffic Manager.	9
IV	Azure Active Directory . Introduction to Active Directory (AD), Identity and Authentication in Public Cloud – Introduction to Azure AD – Extending Active Directory into Azure – Azure AD and applications – Reporting and Monitoring Azure AD – Multi Factor Authentication – Messaging –Biz Talk Services, Media Services . Azure Cost Management and Optimization- Azure Cost Management Tools - Budgeting and Forecasting in Azure - Cost Optimization Strategies	9
V	Azure Databases SQL Azure: Creating a SQL Server - Creating a SQL DB - Creating Tables - Adding Data to the Table - View Connection Strings - Security Configurations - Migrating on premise DB to SQL Azure. Azure Websites: Creating a Website, Setting deployment credentials -Choosing a platform -Setting up Default page for website - Scaling - Auto Scaling by Time -Auto Scaling by Metric - Difference between Free, Shared, Basic and Standard websites - Creating a website using Visual studio.	9

Text Books (TB):

TB-1	Fundamentals of Azure – Microsoft Azure Essentials, Michael Collier and Robin Shahan, Microsoft Press, 2015.
TB-2	Introducing Windows Azure for IT Professionals, Mitch Tulloch with the Windows Azure Team, Microsoft Press, 2013, ISBN: 978-0-7356-8288-7, pp. 1-148.

Reference Books (RB):

RB-1	Implementing Microsoft Azure Infrastructure Solutions, Michael W, Phi Learning Pvt Ltd, 2009.
RB-2	Microsoft Azure Essentials: Azure Automation, Michael McKeown, Microsoft Press, 2015.
RB-3	Microsoft Azure Security Infrastructure, Yuri Diogenes, Microsoft Press, 2016.
RB-4	Cloud Computing with Microsoft Azure, Zainer Tejada, Packt Publishing, 2018.
RB-5	Hands-On Azure for Developers, Kamil Mrzygłód, Packt Publishing, 2020.

Other Reading (OR):

OR-1	Azure Documentation - https://docs.microsoft.com/en-us/azure/
OR-2	Azure Portal - https://portal.azure.com/
OR-3	Microsoft Learn Azure - https://learn.microsoft.com/en-us/azure/
OR-4	Azure Storage Introduction - https://learn.microsoft.com/en-us/azure/storage/common/storage-introduction

Relevant Website (RW):

RW-1	Overview of Microsoft Azure - https://azure.microsoft.com/en-in
RW-2	Azure Virtual Machines - https://learn.microsoft.com/en-us/azure/virtual-machines/
RW-3	Azure Storage Services - https://learn.microsoft.com/en-us/azure/storage/
RW-4	Azure Networking - https://learn.microsoft.com/en-us/azure/networking/
RW-5	Azure Active Directory - https://learn.microsoft.com/en-us/azure/active-directory/
RW-6	Azure SQL Database - https://learn.microsoft.com/en-us/azure/azure-sql/
RW-7	Azure App Service - https://learn.microsoft.com/en-us/azure/app-service/

RW-8	Azure Cost Management - https://learn.microsoft.com/en-us/azure/cost-management-billing/
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Video Links (VL):

VL-1	Introduction to Azure - https://www.youtube.com/watch?v=3Arj5zIUFG4
VL-2	Azure Fundamentals - https://www.youtube.com/watch?v=TJOwP5VhvAo
VL-3	Azure Virtual Machines - https://www.youtube.com/watch?v=inaXkN2UrFE
VL-4	Azure Storage Tutorial - https://www.youtube.com/watch?v=UzTtastcBsk
VL-5	AzureNetworking Basics - https://www.youtube.com/watch?

VL -6	Azure Active Directory - https://www.youtube.com/watch?v=Ma7VAQE7ga4
VL-7	Azure SQL Database - https://www.youtube.com/watch?v=BgvEOkcR0Wk
VL-8	Deploy Web App to Azure - https://www.youtube.com/watch?v=4BwygmRTx8
VL-9	Azure Load Balancer - https://www.youtube.com/watch?v=wJvHwGzvBQ
VL-10	Azure Cost Management - https://www.youtube.com/watch?v=el4yN5cHGS0

MOOC Courses (MC):

Sr.No.	Platform	Topic	CO	Link	Duration
MC-1	Microsoft Learn	Azure Fundamentals	C01	https://learn.microsoft.com/en-us/training/paths/azure-fundamentals/	1hour 30 min
MC-2	LinkedIn Learning	Learning Cloud Computing: Core Concepts	C01	https://www.linkedin.com/learning/learning-cloud-computing-core-concepts	1 hour 23 min
MC-3	Coursera	Introduction to Microsoft Azure Cloud Services	C01 C02	https://www.coursera.org/learn/microsoft-azure-cloud-services	5 hour
MC-4	Microsoft Learn	Configure and Manage Virtual Networks for Azure Administrators	C02 3CO 4	https://learn.microsoft.com/en-us/training/paths/azure-administrator-manage-virtual-networks/	1 hour 40 min

MC-5	Linked In Learning	Azure: Understanding	CO1 CO2	https://www.linkedin.com/learning/azure-understanding-the-big-picture-24392017	2 hour 29 min
MC-6	Microsoft Learn	Implement Azure Security	CO4 CO5	https://learn.microsoft.com/en-us/training/paths/implement-azure-security/	1 hour 5 min
Total MOOC Course integration with Certification					11 hours 55 minutes

Session-wise Planning:

Module	Session	Topic	Pedagogy / Activity planned	CO	Mode of delivery
I	1	Introduction to Windows Azure Web Services, Technologies and TLEP	P1	CO1	M1
	2	Overview of Cloud Computing - Benefits and Challenges	P1		M1
	3	Various Cloud Offerings - IaaS, PaaS, SaaS	P1		M1
	4	Azure Basics - Azure Services Overview	P1		M1
	5	Azure Portals - Preview Portal, Management Portal	P1		M4
	6	Subscription Management - Billing and Pricing Calculator	P1, A1		M1
		Virtual Machine Basics - Status,			

	7	IP Address	P1, A1		M1
	8	Creating and Configuring Virtual Machines	P1		M1,M3
	9	Configuring VM disks and Virtual Machine Management	P1	CO1 CO3	M1,M3
II	10	Storage Basics - Storage Types and Azure Storage Offerings	P2	CO2	M3
	11	Understanding Azure Regions - Using Storage Accounts	P2		M1
	12	Enabling Larger and Faster Storage - Resizing Azure Disks	P2		M1
	13	Using Premium Storage	P2		M1
	14	Monitoring Azure Storage Accounts	P2		M1
	15	Best Practices for Azure Storage	P1		M1

	16	Azure VM Storage Types	P2	CO 2	M3
	17	Azure Files - Managing Azure Storage	P2		M3
	18	Hands-on: Azure Storage Implementation	P2,P3	CO 1 CO 3 CO 4	M4,M2

III	19	Basics of Virtual Networks - Address Spaces, Subnets	P1		M1
	20	DNS Servers in Azure	P2		M1
	21	Creating and Using Virtual Networks	P2		M1
	22	Network Security Groups	P2		M1
	23	Virtual Appliances	P2		M1
	24	Load Balancer basic	P2		M1
	25	Configuring Load Balancers	P1		M1
	26	Azure VPN - Windows Azure Network Services	P1		M1
	27	Traffic Manager - Using Traffic Manager	P1,P3	CO3	M4,M2

IV	28	Introduction to Active Directory (AD)	P2	CO4	M1
	29	Identity and Authentication in Public Cloud	P2		M1
	30	Introduction to Azure AD	P2		M1
	31	Extending Active Directory into Azure	P2		M1
	32	Azure AD and applications	P2		M1
	33	Reporting and Monitoring Azure AD	P2		M3
	34	Multi Factor Authentication	P2		M1
	35	Azure Cost Management Tools	P2		M1
	36	Budgeting and Forecasting in Azure - Cost Optimization Strategies	P1,P3		M4,M2

V	37	SQL Azure: Creating a SQL Server	P2	CO5	M1
	38	Creating a SQL DB - Creating Tables	P2		M1
	39	Adding Data to the Table - View Connection Strings	P2		M1
	40	Security Configurations - Migrating on premise DB to SQL Azure	P2		M1
	41	Azure Websites: Creating a Website, Setting deployment credentials	P2		M1
	42	Choosing a platform - Setting up Default page for website	P2		M1
	43	Scaling - Auto Scaling by Time and Metric	P1		M1
	44	Difference between Free, Shared, Basic and Standard websites	P1		M1
	45	Creating a website using Visual studio -	P2,P3		M4,M2

	Case Study			
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Abbreviations & Expansions					
Pedagogy/Activity planned			Mode of Delivery		
P1	PPT presentation & Classroom interaction		M1	Synchronous – PPT	
P2	Blended Learning with Hands on		M2	Asynchronous	
P3	Flip Class & Module Quiz		M3	Synchronous - Hands on	
A1	Activity-1		M4	Synchronous - Discussion	
A2	Activity-2				

Activity 1: Coursera/Microsoft Learn Courses/NPTEL SWAYAM Courses/ Real Time Projects During live Class Coupled with Faculty-Guided Assessments (10 Marks)				
Criteria	Poor (2-3)	Fair (4-5)	Good (6-7)	Excellent (8-10)
Certificate Completion	Completed only 1 course	Completed 2 courses	Completed 3 courses	Completed 3 courses with high engagement
Course Duration	<15 hrs in 1-2 days	15 hrs in 2-3 days	15 hrs in 4-5 days	15 hrs in 6-8 days
Report Submission	Poor clarity, disorganized	Moderate clarity	Good clarity and structured report	Excellent clarity and well-organized
Oral Presentation & Q&A	Poor voice, language, content	Average communication	Good communication	Very good voice, content & clarity
Activity 2: Azure Mini Project - Identify Application, Specify Requirements, Report & PPT , Real Time Projects During live Class(10 Marks)				
Criteria	Poor (1-4)	Fair (5-6)	Good (7-8)	Excellent (9-10)
Problem Identification	Vague or irrelevant application	Somewhat relevant	Relevant and feasible	Strong real-time

				relevance and clarity
Requirement Specification	Incomplete or unclear	Partial functional/non-functional	Mostly complete	Complete and well-defined
Presentation Clarity	Poor structure & visual aids	Basic structure	Good flow with adequate content	Highly engaging and structured presentation

Assessment Scheme: IA: UE – 50:50

Sl. No.	Assessment Instrument	Formative / Summative	Frequency	Weight age (%)	CO
1	Activity-1 -MOOC Courses	Formative	1	15	CO1, CO2, CO3, CO4, CO5
2	Activity-2 -Mini Project-Case base study		1	15	CO3, CO4, CO5
3	Class participation		-	5	-
4	Internal Exam- 1		1	7.5	CO1, CO2
5	Internal Exam- 2		1	7.5	CO3, CO4, CO5
6	University Exam	Summative	1	50	CO1, CO2, CO3, CO4, CO5
	100				

Bloom's Taxonomy-Revised

LEVEL	DESCRIPTION	MEANING	ACTION VERBS
6	Creating	Can the student create a new product or POV?	Assemble, construct, create, change, combine, compose, design, develop, formulate, invent, modify, organize, propose, theorize, write
5	Evaluating	Can the student justify a stand or decision?	Appraise, agree, assess, argue, conclude, decide, defend, judge, prioritize, prove, rate, recommend, select, support, value
4	Analyzing	Can the student distinguish between different parts?	Contrast, compare, criticize, differentiate, discriminate, dissect, distinguish, examine, experiment, operate, question, simplify, test
3	Applying	Can the student use information in a new way?	Choose, demonstrate, dramatize, employ, illustrate, interpret, schedule, sketch, solve, use
2	Understanding	Can the student explain ideas and concepts?	Classify, describe, discuss, explain, identify, infer, locate, outline, paraphrase, recognize, report, summarize, select, translate
1	Remembering	Can the student recall or remember information?	Define, duplicate, find, list, label, match, memorize, name, omit, recall, repeat, state, spell, tell